HITIMANA JEAN MARIE VIANNEY

GIS & Geospatial AI Engineer

Kigali, Rwanda | +250 789 492 645 | jmvhitimana050@gmail.com | LinkedIn

PROFESSIONAL SUMMARY

Surveying and Geomatics Engineer specializing in drone photogrammetry, GIS, and remote sensing, with a growing specialization in applying computer vision and deep learning to geospatial data. Builds end-to-end AI pipelines from UAV image acquisition through model training to georeferenced GIS outputs and deployed web dashboards for infrastructure monitoring, building change detection, and road defect assessment. Combines strong GIS tooling (ArcGIS Pro, Agisoft Metashape, Global Mapper) with practical software engineering to turn geospatial datasets into automated, production-ready intelligence tools.

KEY AI / GEOSPATIAL PROJECTS

Building Change Detection Pipeline

- Designed and built an end-to-end pipeline to detect building changes (new construction, modifications) between paired drone orthophoto epochs across Rwanda/East Africa imagery, generalizable to a wide range of image resolutions worldwide.
- Implemented tiling, mask vectorization, multi-criteria polygon matching, and three-way change classification, exporting results directly to GIS-ready formats.
- Deployed the pipeline locally with a Flask-based dashboard, giving non-technical users a simple interface to run detections.

AI Pothole Detection & Analysis Pipeline

- Built a computer vision pipeline to detect and size potholes from high-resolution drone orthomosaics, exporting results as GIS layers for direct use in GIS.
- Developed a Flask web dashboard with upload progress tracking and a 2D/3D terrain visualization for inspecting defects in context.

AI Road Defect Detection — Proof of Concept

- Trained detection and segmentation models to identify and classify road defects by severity, producing georeferenced shapefile outputs.

PROFESSIONAL EXPERIENCE

GIS Engineer — *Charis UAS, Kigali* Feb 2025 – Present

- Design and deploy AI/computer vision pipelines for building change detection and infrastructure defect analysis, from model training through to GIS export and dashboard deployment.
- Process UAV imagery using Agisoft Metashape Pro and Pix4D to generate orthomosaics, DTMs, and dense point clouds.
- Conduct large-scale LiDAR corridor mapping and point-cloud/3D processing for infrastructure and cross-border construction monitoring projects.
- Perform spatial analysis and remote sensing workflows in ArcGIS Pro using Landsat and Sentinel-2 imagery.
- Annotate training datasets using CVAT and manage the full ML lifecycle for geospatial computer vision models.
- Collaborate with technical teams to optimize large geospatial dataset workflows, improving processing efficiency and data quality.

Intern, Infrastructure One Stop Center — *Nyanza District* 2023

- Prepared cadastral plans and verified land parcel boundaries.
- Supported land subdivision and ownership verification processes.

- Managed geospatial documentation and land administration records, and assisted clients with land-related submissions.

GIS Tutor & Surveyor (Part-time) — *Muhizi Designing and Construction Ltd* 2022 – 2024

- Delivered GIS training using ArcGIS and ERDAS IMAGINE to students and professionals.
- Supported topographic surveys and spatial data collection.
- Assisted in construction layout design and road alignment surveys using AutoCAD Civil 3D.

EDUCATION

MSc, Environmental and Development Studies — Environmental Information Systems track (In Progress)
University of Lay Adventists of Kigali (UNILAK)

BSc, Surveying and Geomatics (Graduated 2024)
University of Rwanda – College of Science and Technology

Advanced Level Certificate – Construction (2018)
Runda Technical Secondary School

TECHNICAL SKILLS

AI / Computer Vision: Object detection & segmentation models, CVAT annotation, model training & evaluation

GIS & Spatial Analysis: ArcGIS Pro, QGIS, Global Mapper

Remote Sensing: ERDAS IMAGINE, Landsat & Sentinel-2 image analysis

Drone & Photogrammetry: Agisoft Metashape Pro, Pix4D, LiDAR processing, point-cloud/3D mapping

Surveying Tools: GNSS/GPS, Total Station, AutoCAD Civil 3D

Programming & Deployment: Python (geospatial analysis, ML pipelines), Flask, Waitress

LANGUAGES

- Kinyarwanda — Native
- English — Fluent
- French — Basic

REFERENCES

Hakizimana Theogene — GIS Head of Department, Charis UAS | +250 783 344 892

Uwimana Papias — CEO, Muhizi Designing and Construction Ltd | +250 780 620 735 | muhizidesigningacademy@gmail.com

Habimana Evariste — Director, Nyanza Infrastructure One Stop Center | +250 788 660 969